Fill in **just your UID** (fill the bubbles & write the digits above). Each question may have multiple correct answers, **choose all that apply**. Ignore `NULL`s for questions on page 1.

For any relation R the notation $|R|$ means the size of R, i.e., the number of rows in R. For some predicate p, let $r = |R|$, $s = |$`SELECT * FROM R WHERE p`$|$. What are the possible relationships between r and s?

a. $r < s$
b. $r = s$
c. $r > s$

Let x be a column in R. What are the possible relationships between $r = |R|$ and $s = |$`SELECT x from R`$|$?

a. $r < s$
b. $r = s$
c. $r > s$

Let x, y be columns in R. What are the possible relationships between $d = |$`SELECT DISTINCT x FROM R`$|$ and $c = |$`SELECT x, avg(y) FROM R GROUP BY x`$|$?

a. $d < c$
b. $d = c$
c. $d > c$

Let x be a column in R and y be a column in S. Which statement below is **always true** for $r = |R|$, $s = |S|$, $j = |$`SELECT * FROM R, S WHERE R.x = S.y`$|$?

a. $j \leq r + s$
b. $j \leq r \times s$
c. $j \geq \min(r, s)$
d. $j \leq \max(r, s)$

For the same query as above, which statement below **can be true**?

a. $j \leq r + s$
b. $j \leq r \times s$
c. $j \geq \min(r, s)$
d. $j \leq \max(r, s)$

Consider the following queries:

```
J = SELECT R.x FROM R, S WHERE R.y = S.y
E = SELECT R.x FROM R WHERE EXISTS (SELECT * FROM S WHERE R.y = S.y)
```

What are the possible relationships between $|J|$ and $|E|$?

a. $|J| < |E|$
b. $|J| = |E|$
c. $|J| > |E|$

R is a relation that contains no NULLs. Let $I =$ `SELECT * FROM R as r1, R as r2 WHERE r1.x = r2.x` and let $O =$ `SELECT * FROM R as r1 LEFT OUTER JOIN R as r2 ON r1.x = r2.x`. What are the possible relationships between $|I|$ and $|O|$? Choose all that apply.

a. $|I| < |O|$
b. $|I| = |O|$
c. $|I| > |O|$

What is the value of `(NULL + 3) < 4 OR NULL >= 1`?

a. `TRUE`
b. `FALSE`
c. `NULL`
d. `UNKNOWN`

What is the value of `NOT (((NULL * NULL) / NULL) > NULL) AND ((NULL + NULL) <> NULL) OR NOT (NULL < NULL)`?

a. `TRUE`
b. `FALSE`
c. `NULL`
d. `UNKNOWN`

Suppose k is a primary key in R. Let $G =$ `SELECT * FROM R GROUP BY k HAVING count(*) = 1`. What are the possible relationships between $|G|$ and $|R|$?

a. $|G| < |R|$
b. $|G| = |R|$
c. $|G| > |R|$

Suppose k is a primary key in R, and y is another attribute. Let $p =$ `SELECT count(distinct k) FROM R`, and $r =$ `SELECT count(distinct y) FROM R`. What are the possible relationships between p and r? Choose all that apply.

a. $p < r$
b. $p = r$
c. $p > r$

Suppose f is a foreign key in S referencing a primary key k in R. Let $s =$ `SELECT count(distinct f) FROM S` and let $r =$ `SELECT count(distinct k) FROM R`. What are the possible relationships between s and r? Choose all that apply.

a. $s < r$
b. $s = r$
c. $s > r$